

Social Media & Gaming: Predicting Daily Mental Health Outcomes Sana, Ojas, Nik | DTSC Final Project | Spring 2026

1. Research Question & Motivation

Research Question: How accurately can hours of social media and gaming usage, combined with self-reported psychological responses, predict the day-to-day mental health outcomes (daily mood) of individuals?

Mental health is something I've personally seen become more talked about, but still not fully understood, especially among college students. At UNC Charlotte, I've noticed how common it is for students (including myself sometimes) to spend hours on social media or gaming either to relax or escape stress.

What stood out to me is that the impact of this usage is not the same for everyone. Some people feel better after using these platforms, while others feel more anxious, lonely, or drained. That difference is what made this project meaningful to me.

Research also supports this complexity. For example, the Pew Research Center has reported that social media can increase pressure and anxiety among teens, while sources like Psychology Today highlight that online platforms can also provide connection and support. Similarly, reports summarized by Engadget indicate that around 37% of teens report sleep disruption from platforms like TikTok.

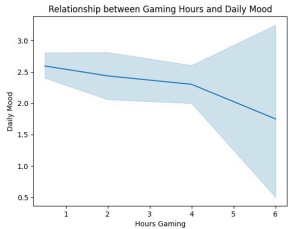
Because of these mixed effects, we wanted to move beyond simple assumptions and use machine learning to detect patterns in how behavior and psychological responses together relate to mood.

2. Dataset & Preprocessing

2.1 Data Source

We collected our own dataset using a Google Form survey distributed to UNC Charlotte students and the surrounding Charlotte, NC community in April 2026. The final dataset contains 100 rows and 28 columns before preprocessing, reduced to 70 usable rows and 11 feature columns after cleaning. Each row represents one individual respondent

2.2 Key Variables

Feature	Type	Description
Social Media Hours	Ordinal	Hours/day on social media (Less than 1 to More than 5)
Video Game Hours	Ordinal	Hours/day gaming (Less than 1 to More than 5)
		
Anxiety Levels	Likert 1-5	Anxiety felt after social media use
Comparison	Likert 1-5	Tendency to compare self to others online
Self Esteem	Likert 1-5	Whether social media affects self-esteem
Using for Relief	Likert 1-5	Using screens to escape stress
Loneliness	Likert 1-5	Feeling lonely despite social media use
Sleep Affected	Likert 1-5	Sleep disruption due to screens
Addiction Levels	Likert 1-5	Self-reported addiction feeling
Gamer?	Categorical	Whether the respondent plays video games

Daily Mood Ordinal Very low, low, Neutral, Good, Very Good
(target)

2.3 Preprocessing Steps

Step 1 - Column removal: We dropped 17 columns that were either free-text responses (e.g., "How does social media affect your mood?"), redundant, or not useful for numerical modeling (e.g., Timestamp, platform selection).

```
Data Janitor!!

# Removing the unnecessary columns
df = df.drop(columns=['How does social media affect your mood?', 'How does gaming affect your stress level?'], axis=1)
df = df.drop(columns=['What do you mostly do on social media?', 'What type of games do you play most?'], axis=1)
df = df.drop(columns=['How often do you interact (likes/comments/replies)?', 'Do you stay responsible for gaming/social media?'], axis=1)
df = df.drop(columns=['Do you check your phone immediately after waking up?', 'Do you play games late at night (past midnight)?'], axis=1)
df = df.drop(columns=['Do you lose track of time while scrolling or gaming?', 'Timestamp'], axis=1)
df = df.drop(columns=['Are you a student?', 'Which platform do you use the most?', 'How often do you check social media?',
                    'Do you play video games?', 'I feel happier after gaming', 'I feel frustrated or angry after gaming',
                    'How many hours of sleep do you get?', 'How often do you feel stressed?'], axis=1)

# Removing the column names
df = df.rename(columns={'How many hours per day do you spend on social media?': 'Social Media Hours',
                    'If you have many hours per day do you game?': 'Video Game Hours',
                    'I feel anxious after using social media': 'Anxiety Levels',
                    'I compare myself to others on social media': 'Comparison',
                    'Social media affects my self-esteem': 'Self Esteem',
                    'I use social media or gaming to escape stress': 'Using for Relief',
                    'I feel lonely even when using social media': 'Loneliness',
                    'My sleep is affected by social media/gaming': 'Sleep Affected',
                    'I feel addicted to social media or gaming': 'Addiction Levels',
                    'On average, how would you rate your daily mood?': 'Daily Mood',
                    'My sleep is affected by social media/gaming': 'Sleep Affected',
                    })
```

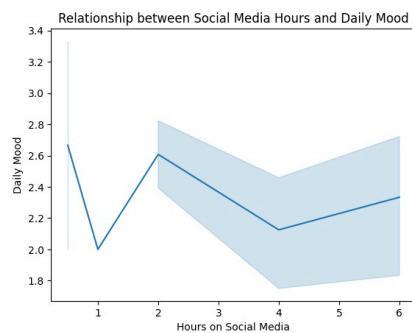
Step 2 - Column renaming: All retained columns were given shorter, readable names (e.g., "On average, how would you rate your daily mood?" became "Daily Mood").

	Age Range	Gender	Social Media Hours	Video Game Hours	Anxiety Levels	Comparison	Self Esteem	Using for Relief	Loneliness	Sleep Affected	Addiction Levels	Daily Mood
0	18-21	Male	3-5 hours	Less than 1 hour	2	2	3	3	3	3	4	2.0
1	22-25	Male	More than 5 hours	Less than 1 hour	4	2	2	2	3	3	2	2.0
2	18-21	Female	3-5 hours	Less than 1 hour	2	1	2	3	1	4	2	3.0
3	18-21	Female	More than 5 hours	3-5 hours	4	5	4	3	3	1	1	3.0
4	18-21	Male	More than 5 hours	3-5 hours	1	2	1	3	2	3	3	3.0

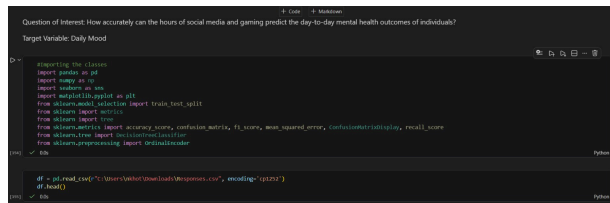
Step 3 - Null value handling: After dropping unnecessary columns, nulls remained in Video Game Hours (29), Using for Relief (3), Loneliness (1), and Sleep Affected (1). Rows with missing Video Game Hours were dropped (non-gamers were marked 0 using ordinal encoding). Remaining nulls in Likert columns were imputed using median values via SimpleImputer during model training.

<pre>df.isnull().sum()</pre> <pre>Age Range 0 Gender 0 Social Media Hours 0 Video Game Hours 0 Anxiety Levels 0 Comparison 0 Self Esteem 0 Using for Relief 0 Loneliness 0 Sleep Affected 0 Addiction Levels 0 Daily Mood 0 dtype: int64</pre>	<pre>df.isnull().sum()</pre> <pre>Age Range 0 Gender 0 Social Media Hours 0 Gamer? 0 Video Game Hours 29 Anxiety Levels 0 Comparison 0 Self Esteem 0 Using for Relief 3 Loneliness 1 Sleep Affected 1 Addiction Levels 0 Daily Mood 0 dtype: int64</pre>
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Step 4 - Ordinal encoding: Social Media Hours and Video Game Hours were encoded using OrdinalEncoder with manually specified category orders to preserve ordinal meaning (e.g., "Less than 1 hour" = 0, "1-3 hours" = 1, "3-5 hours" = 2, "More than 5 hours" = 3). Daily Mood was encoded similarly (Very low = 0, low = 1, Neutral = 2, Good = 3, Very Good = 4).



Step 5 - Target variable binning: For classification models, Daily Mood was collapsed into three classes: "Bad" (0-1), "Neutral" (2), and "Good" (3-4). This was done because the original five-class distribution was too sparse for reliable classification on 70 samples.



```
Question of Interest: How accurately can the hours of social media and gaming predict the day-to-day mental health outcomes of individuals?  
Target Variable: Daily Mood  
  
# Importing the classes  
import pandas as pd  
import numpy as np  
import seaborn as sns  
import matplotlib.pyplot as plt  
from sklearn.model_selection import train_test_split  
from sklearn import svm  
from sklearn import metrics  
from sklearn.metrics import accuracy_score, confusion_matrix, f1_score, mean_squared_error, confusion_matrix, recall_score  
from sklearn.linear_model import LogisticRegression  
from sklearn.preprocessing import StandardScaler  
  
# Load the dataset  
df = pd.read_csv('C:/Users/abhishek/Desktop/Depression.csv', encoding='utf8')  
df.head()
```

Step 6 - Train/test split: An 80/20 split was used, resulting in 56 training samples and 14 test samples.

Assumption: We assumed self-reported responses are approximately accurate and that the distribution of respondents (predominantly 18-25 year old UNC Charlotte students) is representative enough for internal model comparison, though not for external generalization.

3. Preliminary Exploratory Analysis

Before modeling, we created two line graphs: one showing the relationship between social media hours and average daily mood, and another for gaming hours and daily mood. These visualizations revealed a generally downward trend, meaning individuals reporting more screen time tended to report lower average moods, though with notable variance. A countplot confirmed class imbalance: "Good" was the most common mood label, with "Bad" being rare, which informed our choice to use `class_weight='balanced'` in SVM.

4. Model Experiments

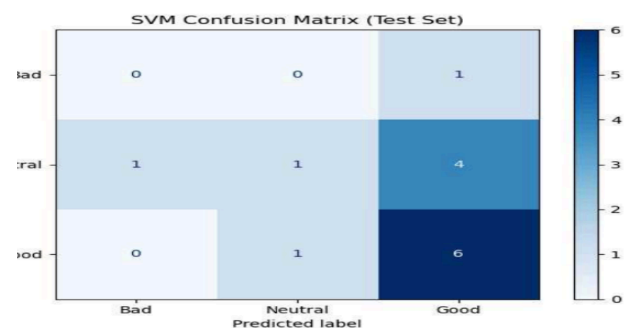
4.1 Model 1: Support Vector Machine (SVM)

Why SVM: SVMs are well-suited to small datasets and work by finding the maximum-margin hyperplane that best separates classes. In our context, the hard margin separates behavioral profiles associated with different mood outcomes. We used the default RBF (radial basis function) kernel, which handles non-linear boundaries, and set `class_weight='balanced'` to compensate for the imbalanced class distribution.

The math: SVM solves the optimization problem: minimize $\|w\|^2 / 2$ subject to $y_i(w \cdot x_i + b) \geq 1$ for all i . With `class_weight='balanced'`, each class gets a weight inversely proportional to its frequency, so the model does not simply predict the majority class.

Results:

Metric	Bad	Neutral	Good	Accuracy
Precision	0.00	0.50	0.55	
Recall	0.00	0.17	0.86	
F1-Score	0.00	0.25	0.67	
Support	1	6	7	
Overall				50.0%



Interpretation: The SVM achieved 50% test accuracy vs 62.5% train, indicating some overfitting. It performed well on the "Good" class (recall 0.86) but failed entirely on "Bad" (recall 0.00), likely due to having only 1 "Bad" sample in the test set. The gap between train and test accuracy suggests the model has memorized some training patterns.

4.2 Model 2: K-Nearest Neighbors (KNN)

Why KNN: KNN classifies a new sample by looking at the k most similar training examples and taking a majority vote. It makes no assumptions about the distribution of the data, which is valuable when we do not know whether relationships between features are linear or Gaussian. KNN is also intuitive: a person with similar social media

habits and psychological responses to peers with "Good" moods is likely to also have a "Good" mood.

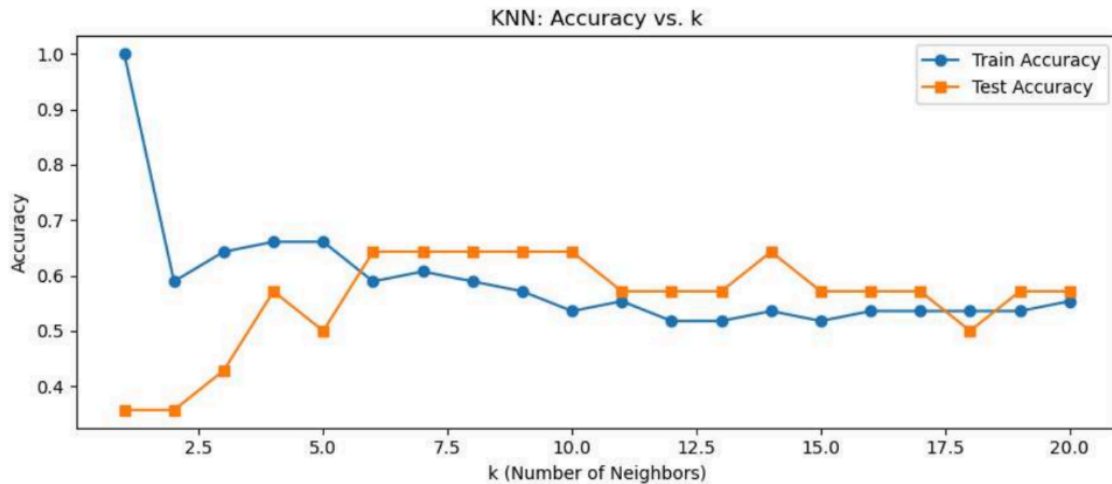
The math: For a new point x , KNN finds the k training points x_i with smallest Euclidean distance $d(x, x_i) = \sqrt{\sum (x_j - x_{i_j})^2}$ and assigns the majority class label. We use StandardScaler to normalize features before computing distances.

Hyperparameter tuning: We tested k values from 1 to 20 and found $k=6$ produced the highest test accuracy. Smaller k values (1-3) overfit the training data; larger values (15+) underfit by averaging over too many dissimilar neighbors.

Results:

Metric	Bad	Neutral	Good	Accuracy
Precision	0.00	1.00	0.58	
Recall	0.00	0.33	1.00	
F1-Score	0.00	0.50	0.74	
Support	1	6	7	
Overall				64.3%

Interpretation: KNN with $k=6$ was our best model at 64.3% test accuracy. It achieved perfect recall (1.00) on the "Good" class, correctly identifying every person with a good mood. The Neutral class had perfect precision (1.00), meaning when it predicted Neutral it was always right, but low recall (0.33), meaning it missed many Neutral cases. Both models failed on "Bad" due to extreme class imbalance in the test set.



Best k on test set: 6 (Test Accuracy = 0.643)

```
knn.fit(X_train, y_train)
```

▼ KNeighborsClassifier ① ②

```
KNeighborsClassifier(n_neighbors=9)
```

4.3 Model 3: Linear Regression (Baseline Comparison)

Why included: Linear Regression was included as a baseline to test whether a simple continuous predictor could capture mood trends. It was not expected to perform well since Daily Mood is ordinal/categorical, not a true continuous variable.

Result: $R^2 = -0.24$. A negative R^2 means the model performs worse than simply predicting the mean every time. This confirms that Linear Regression is the wrong tool for this problem and validates our choice to use classifiers instead. This result is informative because it tells us the relationship between features and mood is not linear and requires a more complex decision boundary.

4.4 Model 4: Probability Calibration (Beyond Class Method)

Why Probability Calibration: Standard classifiers like SVM output a predicted class but not a reliable confidence score. CalibratedClassifierCV from sklearn.calibration wraps a base classifier and maps its raw outputs to well-calibrated probabilities using isotonic regression, a monotone non-parametric fitting approach applied via 5-fold

cross-validation. This was not covered in class and represents the novel method for this project.

Why this matters: In a mental health context, knowing a model's confidence is critical. A prediction of "Bad mood" with 95% probability should be treated very differently from one with 51% probability. Calibration enables this nuance.

Result: The Calibrated SVM matched the base SVM at 50.0% test accuracy. However, it successfully produced probability scores for each class. For example, one test sample received: Bad: 0.028, Good: 0.536, Neutral: 0.436, indicating near-equal uncertainty between Good and Neutral. This probabilistic output is more informative than a hard label for real-world use cases.

4.5 Experiment: Feature Selection with Decision Tree

Why Decision Tree: Decision Trees inherently rank features by how much they reduce Gini impurity at each split, providing a natural importance score. We used `max_depth=4` to prevent overfitting.

Experiment A - All Features: Test accuracy = 57.1%. The tree used all 10 features and revealed that psychological variables (Anxiety Levels, Addiction Levels, Loneliness, Sleep Affected) ranked higher in importance than raw screen-time hours alone.

Experiment B - Selected Features (Social Media Hours, Video Game Hours, Anxiety Levels, Sleep Affected, Addiction Levels): Test accuracy = 42.9%. Removing demographic and other psychological features reduced performance, confirming those features do add predictive signal.

Conclusion: Screen-time hours alone are insufficient predictors. The psychological side-effects of technology use, especially anxiety, sleep disruption, and addiction feelings, are more predictive of daily mood than hours logged.

5. Model Comparison Summary

Model	Train Accuracy	Test Accuracy	Note
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KNN (k=6)	58.9%	64.3%	Best overall
DT (All Features)	76.8%	57.1%	Reveals feature importance
SVM	62.5%	50.0%	Class-balanced
Calibrated SVM	67.9%	50.0%	Adds probabilities
DT (Selected)	62.5%	42.9%	Screen-time only
Linear Regression	N/A	-0.24 R ²	Baseline only; wrong tool

KNN with k=6 performed best on the test set at 64.3%, with a train accuracy of 58.9%, the closest train-test gap of any model, indicating the least overfitting. The Decision Tree with all features overfits most severely (76.8% train vs 57.1% test). The consistent failure to predict the "Bad" mood class across all models is a limitation tied to extreme class imbalance in the test set (only 1 "Bad" sample out of 14).

6. Limitations, Ethics & Reflection

6.1 Limitations

- **Small, geographically narrow sample:** About 70 usable rows, almost entirely UNC Charlotte students in Charlotte, NC. Results cannot be generalized nationally or across age groups.
- **Self-reported data:** Respondents may inaccurately recall screen time or rate their mood based on how they feel at the moment of filling out the survey.
- **Class imbalance:** The "Bad" mood class had very few examples. Despite `class_weight='balanced'` in SVM, no model successfully predicted a "Bad" outcome in the test set due to only 1 Bad sample among 14 test points.
- **Train/test split instability:** With only 14 test samples, a single misclassification changes accuracy by about 7%. These results have high variance and should be interpreted cautiously.

- **Linear Regression mismatch:** Daily Mood is ordinal/categorical. Its negative R^2 confirms it is unsuitable and was included for experimental comparison only.
- **Causality:** Our models identify correlation, not causation. A third factor such as academic stress could independently explain both high screen time and low mood.

6.2 Ethical Considerations

- **Privacy:** All responses were anonymous. No personally identifiable information was collected.
- **Stigma risk:** Labeling mood states from behavioral data could reinforce stigma. This model must never be used to screen, gatekeep, or make decisions about individuals.
- **Bias:** The sample skews toward 18-25 year old college students. Predictions for older adults, non-students, or different cultural contexts may be systematically wrong.
- **Not a clinical tool:** This project is strictly for academic awareness. Any prediction here must not substitute for professional mental health assessment.

6.3 Reflection

We were surprised that KNN outperformed SVM given SVM's theoretical advantages on small datasets. The feature importance experiment was the most revealing part of the project. It shifted our interpretation from "screen time predicts mood" to "the psychological relationship with technology predicts mood." If we had more data, we would explore SMOTE oversampling for the Bad mood class, cross-validation, and possibly a Random Forest or Gradient Boosting approach. We would also collect data from a broader geographic and demographic base.

7. References & AI Transparency

Dataset:

- Survey data self-collected via Google Forms by Sana, Ojas, and Nik (April 2026). Distributed to UNC Charlotte students and Charlotte, NC community. About 100 responses.

Libraries:

- pandas, numpy - data loading and manipulation
- matplotlib, seaborn - visualizations
- scikit-learn - OrdinalEncoder, train_test_split, SVC, KNeighborsClassifier, LinearRegression, DecisionTreeClassifier, CalibratedClassifierCV, SimpleImputer, classification_report, confusion_matrix, ConfusionMatrixDisplay

External Sources:

- Psychology Today - social media and community connection benefits
- Engadget - 9% of teens negatively impacted; 37% sleep affected by TikTok
- Open Textbooks - cognitive benefits of video games
- Pew Research Center - teen pressure and social media anxiety
- <https://www.youtube.com/watch?v=SjOfbbf12qY>

AI Use Transparency - Claude (Anthropic, claude-sonnet-4-6) was used to help fix code errors. All code was reviewed and run by the team. Data collection, preprocessing, core model implementations, and survey design were completed independently.